

Mentors, Visits, and Learning: Evidence from Telangana’s FLN Reform

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April 2026

Abstract

This report examines whether school mentor visits under Telangana’s Foundational Literacy and Numeracy (FLN) reform are associated with improvements in student learning outcomes. It combines administrative school data, mentor visit records, and multiple rounds of student assessment data to analyze both patterns in visit delivery and their relationship to learning. Descriptively, visit frequency increased substantially over time, the urban–rural gap in visit coverage narrowed in 2025/26, and score distributions shifted across testing waves. The study then applies two empirical approaches. A school-level baseline-adjusted analysis relates March 2025 performance to cumulative mentor visits, controlling for prior test scores. This approach does not identify a robust positive relationship between visits and learning outcomes. A district-level difference-in-differences analysis comparing DPMU and non-DPMU districts similarly finds no statistically significant relative gains in Math, English, or Telugu. Overall, the findings suggest that increased visit counts alone did not improve learning outcomes; however, data constraints, a short follow-up window, and non-random visit allocation mean these results should be interpreted as suggestive rather than causal.

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1 Introduction

There is growing evidence that children who fail to master basic reading and numeracy in the early grades face lasting disadvantages across education, employment, and health (Das, Andrabi, & Khwaja, 2023; What Works Hub for Global Education, 2026). In Telangana, this prompted the state government to launch a Foundational Literacy and Numeracy (FLN) mission in August 2022, supported by the Central Square Foundation (CSF).

A central but often overlooked lever in education reform is the layer of middle managers responsible for supervising teachers. In Telangana, as in many Indian states, these officials have historically functioned as compliance monitors, focused on administrative checks rather than instructional support (CSF, 2025). There is limited evidence on whether shifting this role toward classroom observation and teacher feedback can improve learning outcomes at scale. CSF’s Strengthening Middle Management pathway aims to drive this shift by supporting mentors to observe classrooms regularly and provide structured feedback to teachers.

This study examines whether this shift is associated with improved student learning outcomes, through three questions. First, we examine how mentor visits are distributed across schools, and whether that distribution is equitable. Second, we examine whether schools receiving more visits achieved better outcomes at endline, controlling for baseline performance. Third, we use variation in CSF support intensity across districts to ask whether more intensively supported districts experienced larger learning gains over time.

The paper proceeds as follows. Section 2 provides background on Telangana’s school system and the FLN reform. Section 3 describes mentor roles and visit expectations. Section 4 sets out the theory of change and study scope. Sections 5 and 6 present the research questions and data. Sections 7 and 8 present the descriptive analysis and empirical strategy. Section 9 discusses the findings and study limitations, Section 10 concludes.

2 Background and Context

This study focuses on Telangana, a state in south-central India formed in 2014 following the bifurcation of the former state of Andhra Pradesh. The School Education Department is the largest government department in the state, aiming to impart general education to all children in the age group of 6 to 15 years. The state has over 41,939 schools, of which over 26,152 are public schools.

2.1 The Foundational Learning Crisis in Telangana

A large-scale assessment conducted in Telangana in 2021 revealed low performance in foundational Mathematics, English, and Telugu. It found that only 43% of students in Grade 5 could read basic texts with comprehension, and only 40% of students in Grade 3 could use basic math operations to solve daily life problems (CSF, 2022). These findings prompted the Department of School Education, Government of Telangana, to launch a Foundational Literacy and Numeracy (FLN) mission on 15 August 2022. In Indian government policy, a mission is a time-bound programme with a defined goal, dedicated resources, and coordinated delivery across administrative levels. The state mission’s goal is to ensure that every child in Grades 1–5 can read and do basic maths at the level expected for their grade.

2.2 The Central Square Foundation’s Role in the FLN Reform

The Central Square Foundation (CSF) partners with the Telangana state government to support the delivery of the FLN mission through reform across five pathways. These pathways consist of the following: Governance and Salience Building, Strengthening Middle Management, Strengthening Data Systems and Data Validation, Academic Inputs, and Community Action (CSF, 2024a).

This study focuses on the Strengthening Middle Management pathway. We examine differences in mentor visit exposure across schools – including across districts with different levels of CSF support – and whether these differences are associated with differential impacts on student learning outcomes. This is carried out with two main empirical approaches which are explained in further detail.

2.3 Administrative Structure and the Role of Mentors

The education system in India is organized hierarchically from state to district to mandal to cluster to school. At the district level, 33 District Education Officers (DEOs) guide and review performance across the system. Below them, 629 Mandal Education Officers (MEOs) operate as mandal-level supervisors, visiting schools, observing lessons, and coaching headmasters and teachers. At the cluster level, 1,509 Complex Headmasters (CHMs) each lead a designated “school complex” of nearby schools, providing first-line instructional support and mentoring. CSF’s reform efforts focus primarily on this cluster level. These officials are referred to as *mentors* throughout this study.

Mentors are not external coaches but part of the existing government system. They are typically experienced headteachers who have been given additional responsibility for supporting other schools in their cluster. This matters for how the role works in practice. On the one hand, they understand the system and the constraints teachers face. On the other hand, they are balancing mentoring with their own school leadership responsibilities. The FLN reform represents a shift in what is expected from them, moving from a more administrative role toward one that focuses on classroom practice and teacher support.

3 Expectations of Mentors

3.1 The Mentoring Role

Mentors provide guidance and support to teachers and headmasters of the schools within their designated complex. This guidance is delivered through school visits, during which the mentor observes a teacher conducting a class and records notes on teaching practices and adherence to state-provided teaching framework guidelines. The mentor then provides feedback to the observed teacher at the end of the class. This feedback is expected to cover classroom management, student engagement, the effectiveness of exercises and activities, and use of teaching-learning materials.

Visits also include spot assessments of students conducted directly in the classroom. Mentors observe teaching against the state’s 5+1 instructional model, which prescribes specific steps by subject. For Telugu and Urdu, these steps are: oral language development, introduction of keywords, model reading, shared reading, individual reading, comprehension activities, and story time. For Mathematics, the steps are: pre-knowledge test, introduction to concept, activities for conceptual understanding, problem solving, recap, and independent student practice (CSF, 2022).

3.2 Visit Frequency Targets and Mentor Autonomy

Ideally, mentors are expected to visit each school in their cluster at least once per month. However, based on consultation with CSF, a goal of six visits per school, per year was set as a realistic target and is considered the benchmark target in this study. Mentors have the autonomy to choose which schools to visit and how frequently. They can plan their visits based on their judgment of perceived need, accessibility, or other contextual factors.

Each mentor is responsible for multiple schools within a cluster, and visits must be planned across these schools over the course of the year. As a result, it may not be possible to visit all schools with the same frequency. Given that mentors can choose which schools to visit and how often, the distribution of visits across schools is not fixed and may vary.

3.3 The Role of DPMU Districts

CSF does not provide equal levels of support across all districts. In a subset of districts designated as District Project Management Unit (DPMU) districts, CSF operates with a higher level of direct engagement compared to non-DPMU districts. The purpose of the DPMU intervention is ensuring implementation fidelity across the reform pathways. This includes additional support towards the delivery of teacher training and capacity building, encouraging wider and more continuous mentor visit coverage, and ensuring that FLN progress reviews take place at all administrative levels on a monthly basis. In practice, mentors in DPMU districts receive dashboards and reports showing which schools in their clusters have been visited, how frequently, and which schools may be receiving insufficient attention, creating structured accountability for both compliance and visit quality throughout the year.

There are five DPMU districts in Telangana. These include: Bhadrachalam, Medak, Medchal Malkajgiri, Narayanpet, and Vikarabad. Treatment in the first three began in November–December 2023, while Narayanpet and Vikarabad were added in September–October 2024. This variation in the intensity of CSF’s support across districts provides a basis for comparing learning trajectories between DPMU and non-DPMU schools over time, which forms the second empirical approach of this study.

3.4 Compliance versus Quality

Mentor performance can be evaluated along two dimensions: compliance, meaning whether mentors meet their expected visit frequency targets, and quality, meaning whether visits deliver meaningful and insightful guidance to teachers. This study focuses exclusively on the compliance dimension. Quality is excluded for several reasons: there are no records of the specific verbal feedback provided by mentors to teachers; the online application’s branching logic produces observation records that are difficult to compare consistently across visits; and CSF staff noted concerns about top-coding, whereby mentors may select the highest response categories to remain diplomatic, further limiting the reliability of recorded quality measures. This study therefore uses the frequency of mentor visits as the primary measure

of mentoring exposure.

4 Theory of Change, Motivation, and Study Scope

4.1 The Theory of Change

According to CSF’s theory of change, the middle management pathway works through a simple chain. Mentors visit schools, observe classrooms, and give feedback to teachers. The idea is that this support helps teachers improve what they do in the classroom. Over time, better teaching should lead to better learning outcomes in literacy and numeracy.

In practice, this depends on more than just how many visits happen. What matters is also what happens during those visits. If feedback is not useful, or if it is too superficial, more visits may not change much. At the same time, visits are not randomly assigned. Mentors may choose schools based on size, location, or ease of access, which can affect who actually receives support.

This study looks at the link between visits and student learning. It asks whether schools that receive more visits see better outcomes, and whether districts with more structured support show larger improvements over time. At the same time, the analysis cannot directly measure changes in teaching practices, so it does not fully capture the middle step in this process.

4.2 Motivation and Study Scope

Despite the centrality of the mentoring pathway in CSF’s TOC, there is limited evidence on how it operates in practice and how consistently it reaches schools. Since mentors have substantial autonomy in deciding which schools to visit and how frequently, mentoring exposure can vary considerably both within and between school complexes, with some schools receiving far more support than others. This uneven implementation could produce uneven learning gains. There is also an important equity concern: if mentors systematically prioritize larger, more accessible, or better-performing schools while neglecting smaller, remote, or disadvantaged schools, the FLN reform could inadvertently widen existing achievement gaps rather than narrow them. CSF currently monitors compliance primarily through aggregate visit targets, which may obscure these distributional differences.

This study addresses these gaps through three complementary lines of analysis. First, we

examine whether mentor visits are systematically distributed across schools. We present descriptive analysis of visit patterns and balance regressions to test whether schools receiving more visits differ in a statistically significant way from those receiving fewer in terms of observable characteristics such as enrollment, location, and student composition. Second, we examine whether schools that received more mentor visits during the 2024/25 academic year achieved better learning outcomes at endline, controlling for baseline performance. Third, we use variation in CSF support intensity across DPMU and non-DPMU districts to ask whether schools in more intensively supported districts experienced larger gains in learning outcomes over time. Together, these approaches provide evidence on both the equity of program implementation and whether greater mentor visit exposure is associated with improved student learning outcomes.

5 Research Questions

This study addresses three sets of research questions, covering descriptive, monitoring, and learning dimensions of CSF’s intervention.

Descriptive Questions

- What is the distribution and frequency of visits to schools?
- How do visit patterns vary by school characteristics?

Monitoring Questions

- Are schools receiving the expected number of visits?

Learning Questions

- Do mentor visits improve student learning outcomes?
- Have DPMU districts experienced larger gains in learning outcomes over time than non-DPMU districts?

6 Data

6.1 Data Sources

This analysis draws on three main data sources. The first is the Unified District Information System for Education Plus (UDISE+) national school census, which provides the sampling frame and school-level administrative characteristics. The second is CSF’s classroom observation records, which serve as the primary measure of mentoring exposure. The third is CSF’s sample-based student assessment data, which provides the main outcome measures.

6.2 UDISE+ Administrative Data

UDISE+ is compiled annually by the Department of School Education and Literacy, Government of India, and is the official national education management information system. It collects standardized administrative information for all recognized schools in India. This study uses the 2025 academic year UDISE+ dataset to construct a census of all government schools in Telangana. From this dataset, we utilize unique school identifiers, teacher counts, precise school location, and enrollment counts disaggregated by grade, gender, and social category (Other Castes, Scheduled Castes, Scheduled Tribes, and Backward Classes). Critically, UDISE+ acts as the sampling frame for the study. Since CSF’s mentor visit records include only schools that were visited, UDISE+ allows identification of schools that received zero mentor visits. This enables analysis of both the extensive margin (whether a school was visited at all) and the intensive margin (how many visits it received).

6.3 Classroom Observation Data

CSF’s classroom observation records are collected at the visit level by mentors during school visits. This dataset provides a record of the date of each visit, the school and classroom observed, and basic class characteristics such as subject and grade. These records are aggregated to the school-year level to construct visit counts per school for the 2024/25 and 2025/26 academic years. These counts are used exclusively as a coverage and exposure measure. The pedagogical content of the observations is not used in the analysis due to concerns about data reliability and the complexity of branching logic in the online instrument.

6.4 Student Assessment Data

CSF carried out sample-based student assessments in the academic years 2022, 2023, 2024 and 2025. In 2022 and 2023, assessments covered a sample of students from all 33 districts. In 2024, samples were collected in only six districts. Assessments were administered at the student level, with item-level binary indicators for correct and incorrect responses across multiple FLN competencies. The competencies assessed include, in Mathematics: Number Identification, Number Identification Understanding, Number Comparison, Addition (Level 2), and Subtraction (Level 2); in English: Word Reading, Oral Reading, Reading Comprehension, and Word Writing; and in Telugu: Word Reading, Oral Reading, Reading Comprehension, Word Writing, and Oral Vocabulary.

6.5 Sample Construction and Filtering

Table 1 documents the baseline filtering steps used to construct the core sample used throughout the paper. Additional filters are applied for the two empirical approaches, as described later. Beginning with 41,939 schools in UDISE+, the sample is filtered to public schools only (26,152). Secondary-only schools covering grades 6–10 are dropped (21,448 remaining) and schools with zero enrollment are excluded (19,572 remaining).

Table 1: Sample Construction

Step	Filter	Schools
1	All schools in UDISE+	41,939
2	Public schools only	26,152
3	Drop secondary-only schools (grades 6–10)	21,448
4	Enrollment > 0	19,572

6.6 Data Availability and Timing

The analysis in this study uses three baseline assessment rounds: August 2022 (Baseline 1), August 2023 (Baseline 2), and September 2024 (Baseline 3). The treatment window, of tracked mentor visits, runs from September 2024 to February 2025. The endline assessment data is from March 2025. This means the regression analysis links visits occurring within a six-month window to endline outcomes measured one month after the treatment ends.

Whilst the descriptive work uses both rounds of mentor visit data (2024/2025 and 2025/2026 academic years), the empirical analysis does not use mentor visits from the 2025/26 academic year. This is because the 2025/26 school year begins in June 2025, which occurs after the March 2025 outcome measurement. Therefore, visits recorded in 2025/26 occur after the endline assessment.

6.7 Approach 1 Outcome Variable Construction: Anderson Index

For Approach 1, we use an Anderson index as our outcome variable. Rather than analyzing each skill component separately, which would require multiple hypothesis tests, the Anderson index aggregates multiple skill measures into a single summary index per subject. This reduces the multiple testing problem while retaining the full information across skill dimensions. The index is constructed separately for each subject by combining multiple skill measures. For Mathematics, these include number identification, number comparison, addition, and subtraction. For English and Telugu, they include word reading, oral reading, and writing.

First, each component is standardized using the August 2023 baseline by subtracting the mean and dividing by the standard deviation, putting all skills on the same scale. Second, the correlation across components is used to construct Anderson weights: components that are highly similar to others receive less weight, while those providing more independent information receive more weight. This inverse covariance weighting ensures that redundant measures do not dominate the index. Third, the standardized components are combined into a single index for each student using these weights. Finally, the combined index is re-standardized using the August 2023 baseline so that it has mean zero and standard deviation one, allowing results to be interpreted in standard deviation units.

7 Descriptive Analysis

This section provides an overview of mentor visit patterns across schools, focusing on the distribution, frequency, and variation in visit intensity. The aim is to assess whether schools are receiving the intended level of support and to highlight any differences in implementation over time and between different schools.

Figures 1 and 2 show that visit frequency remains low overall, though there has been a substantial improvement over time. The median number of visits per school doubled from

2 to 4 between 2024/25 and 2025/26, and the mean increased from 2.81 to 4.52. The share of schools receiving at least six visits, the minimum benchmark, marked by a red line in Figure 1 also increased considerably between the two years. However, even in 2025/26, only around 30% of schools received the required number of visits.

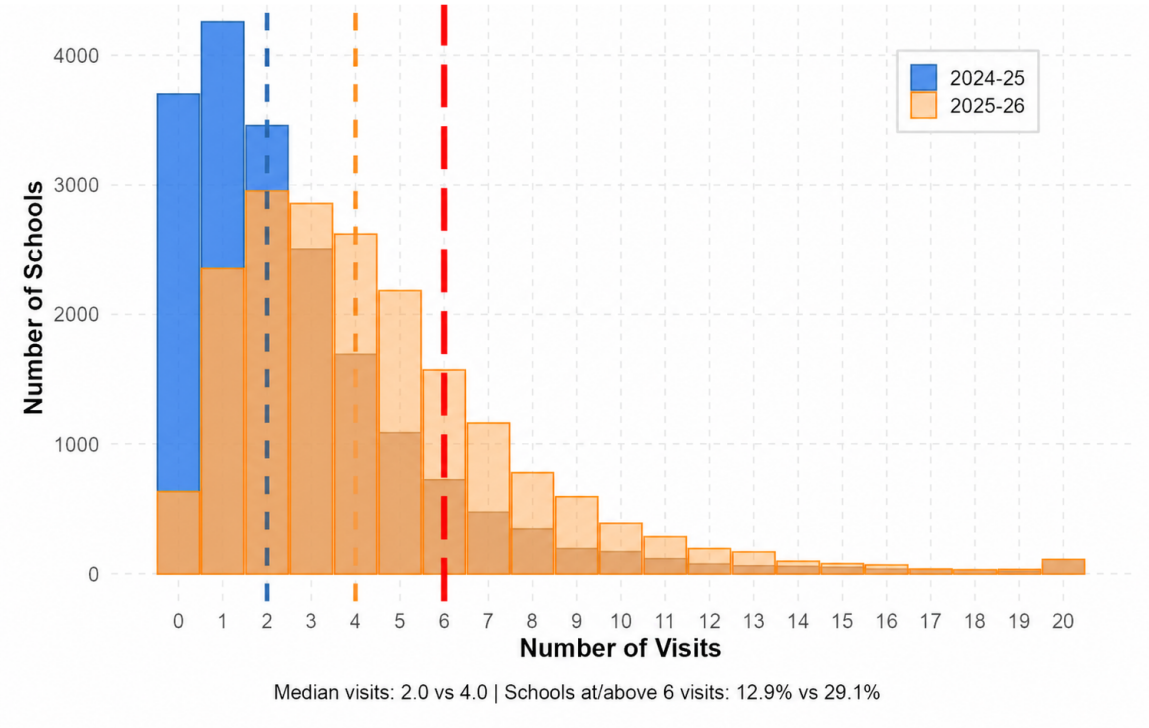


Figure 1: Average number of mentor visits per school, 2024/25 and 2025/26.

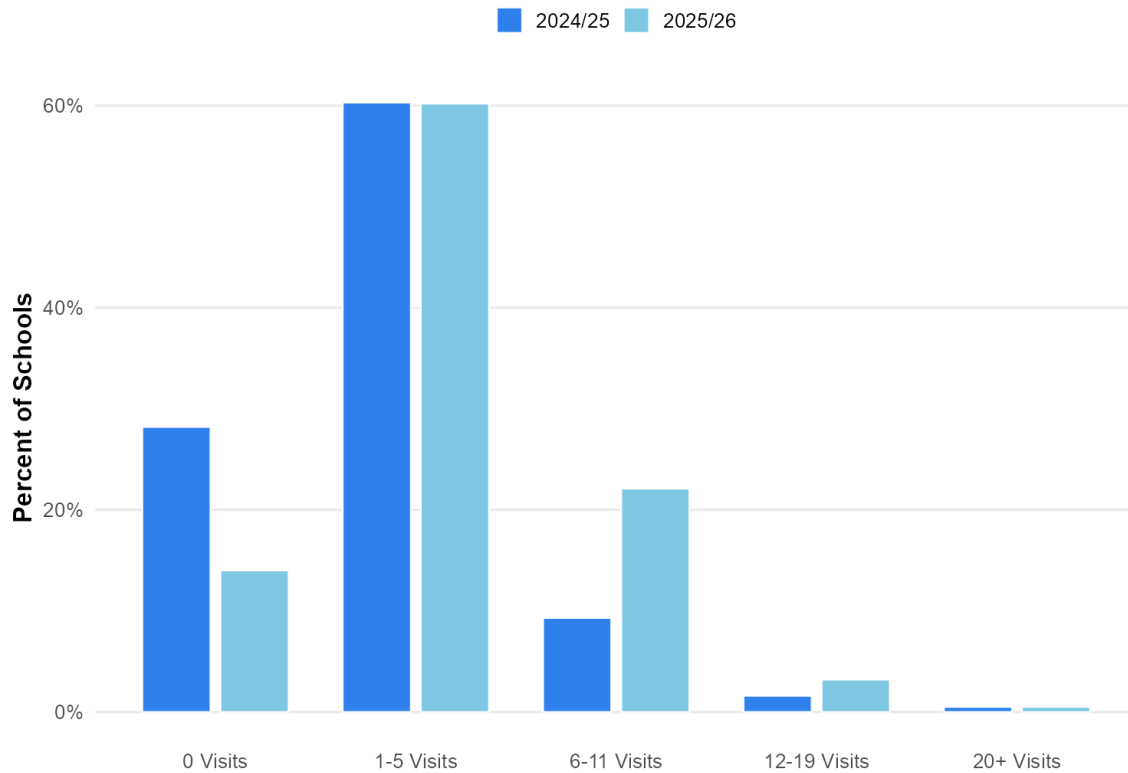


Figure 2: Distribution of schools by number of mentor visits received, 2024/25 and 2025/26.

Beyond overall frequency, Figures 3 and 4 show visit counts against school enrollment disaggregated by urban and rural status, in each year. Visits are higher in larger and more urban schools in 2024/25, and that the urban-rural gap narrows in 2025/26.

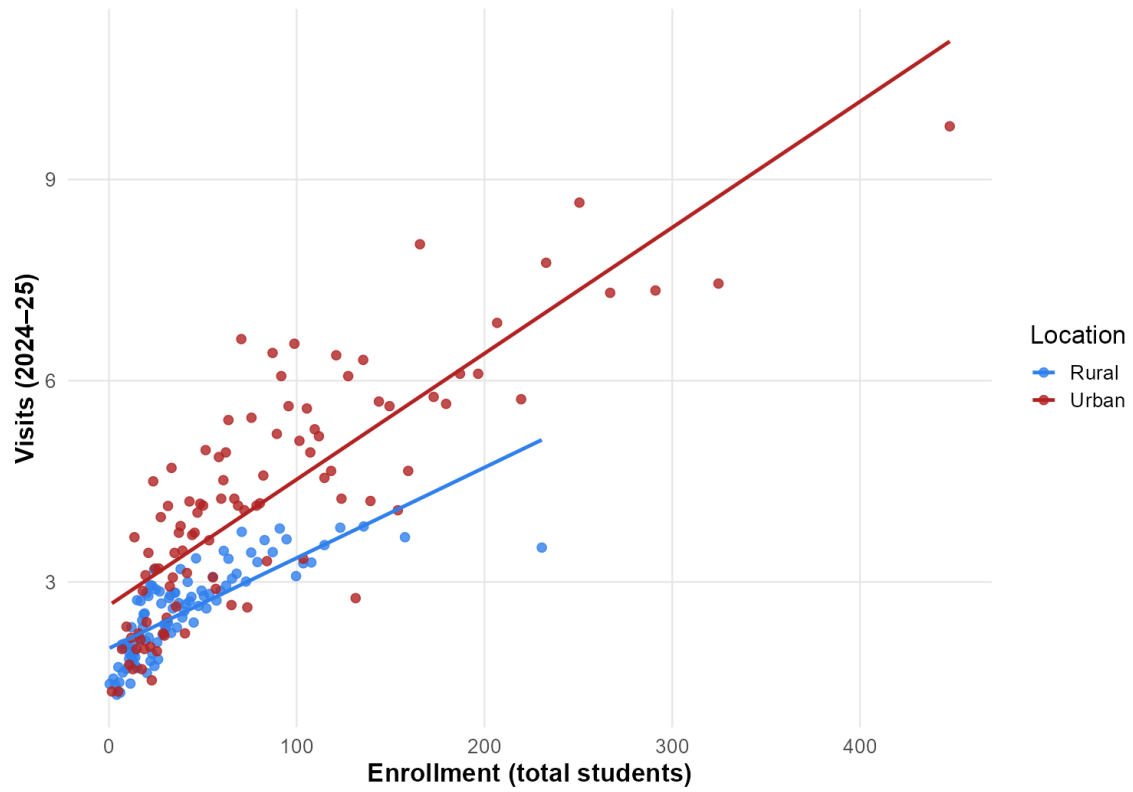


Figure 3: Visits versus enrollment by urban/rural status, 2024/25.

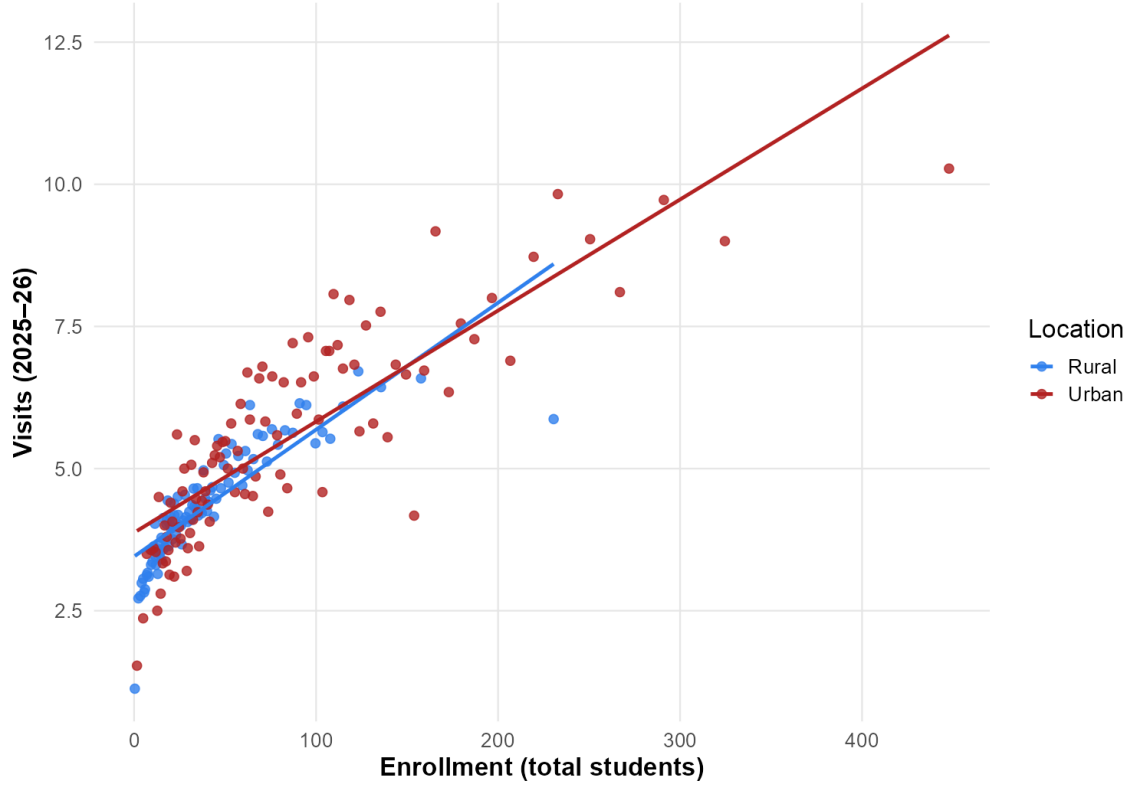


Figure 4: Visits versus enrollment by urban/rural status, 2025/26. The urban–rural gap narrows relative to 2024/25.

Causal Identification Challenges and Balance Checks

A key challenge for approach 1 is that visit allocation is non-random. Mentors are able to exercise discretion in choosing which schools to visit, and the balance checks have shown that schools receiving more visits are systematically larger, more urban, and have a lower proportion of SC/ST students. To assess the severity of this problem formally, school characteristics measured in August 2023 are regressed on visit counts as follows:

$$Y_{s, \text{Aug23 characteristic}} = \beta_0 + \beta_1 \text{Visits}_s + \varepsilon_s \quad (1)$$

The results, reported in Table 2 confirm that schools receiving more visits are systematically larger (coefficient 3.539, significant at the 1% level), more likely to be urban (0.041, significant at the 1% level), and serve lower shares of SC/ST students (−1.407, significant at the 10% level). These results suggest that mentor visit allocation is not random. This means that a simple comparison of high-visit and low-visit schools would conflate any effect of visits with pre-existing differences in school characteristics. Both approaches attempt to

address this through controls and fixed effects. This concern is less applicable to Approach 2, which operates at the district level rather than the school level; since DPMU districts were designated at a higher administrative level, school-level sorting within districts is less likely to drive the estimates. Nonetheless, the results should be interpreted as suggestive associations rather than causal estimates.

Table 2: Balance and Targeting Check: School Characteristics Regressed on Visit Counts

	Enrollment	Urban	SC/ST
Visits	3.539***	0.041***	−1.407*

Notes: Each column reports the coefficient from a separate OLS regression of the school characteristic (measured in August 2023) on visit counts. For SC/ST, each additional visit is associated with a 1.4 percentage point lower share of SC/ST students. * $p < 0.10$, *** $p < 0.01$.

Turning to student learning outcomes, Figures 5, 6, and 7 document score distributions in Mathematics, English, and Telugu across the four assessment waves of August 2022, August 2023, September 2024, and March 2025, as descriptive context for the regression analysis. These distributional shifts should be interpreted with caution: differences across waves may reflect changes in exam difficulty or assessment sampling rather than genuine improvements in student learning. In Mathematics, the distribution shifts steadily to the right over time, with increasing concentration in the upper score range by September 2024 and March 2025. English and Telugu show a different pattern whereby the August 2022 distributions are more concentrated in the lower-to-middle score range, while later waves are more evenly spread across the distribution, with less clustering at low scores. By 2024 and 2025, all three subjects exhibit fewer very low scores and greater mass in the middle-to-high score ranges, although the shift is strongest in Mathematics.

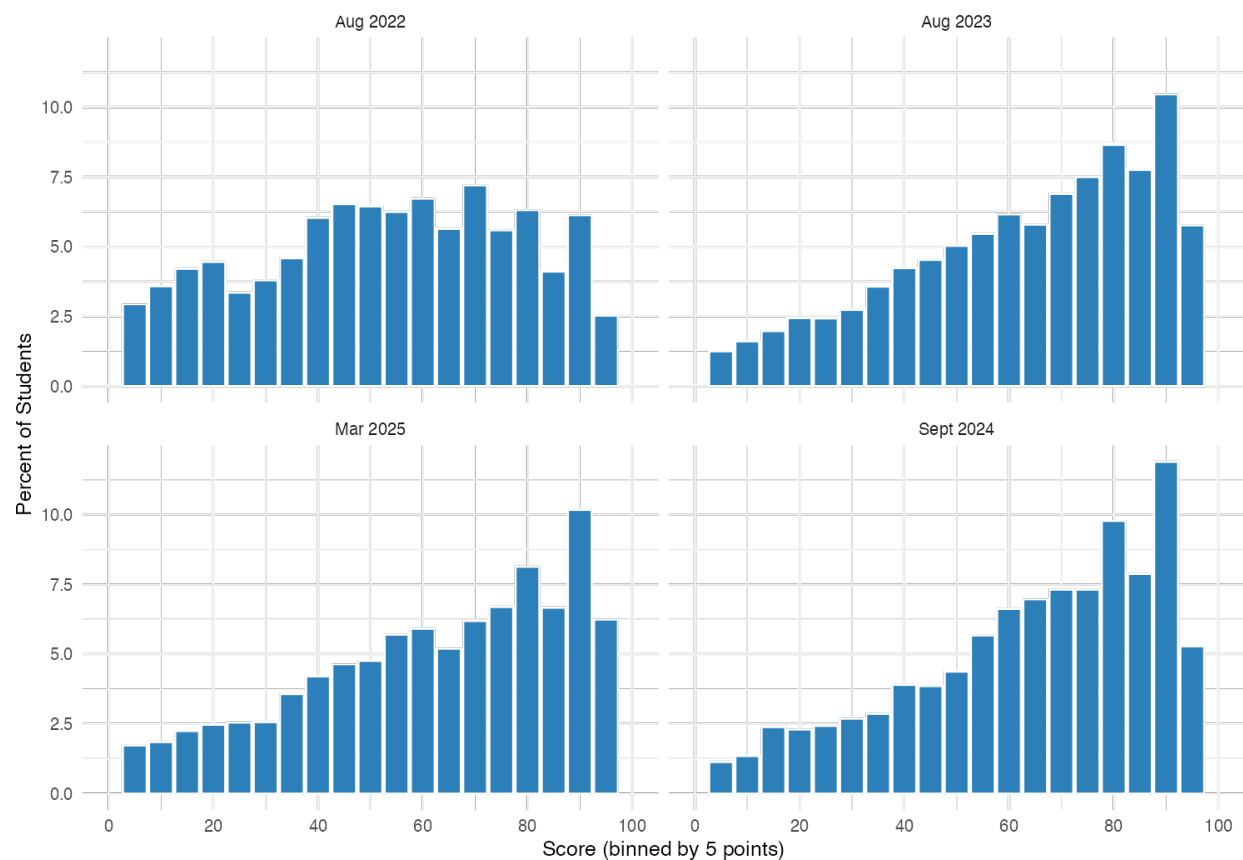


Figure 5: Distribution of school-level Mathematics scores across assessment waves.

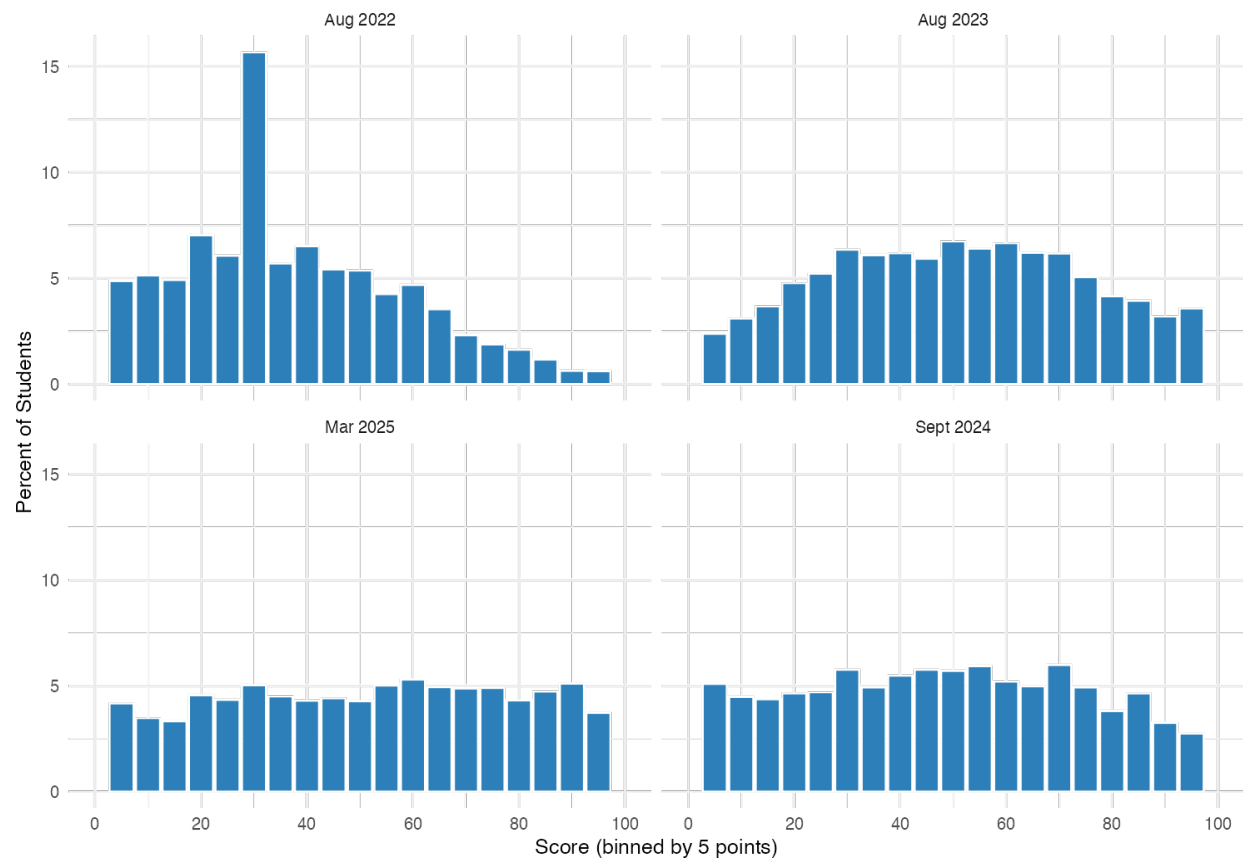


Figure 6: Distribution of school-level English scores across assessment waves.

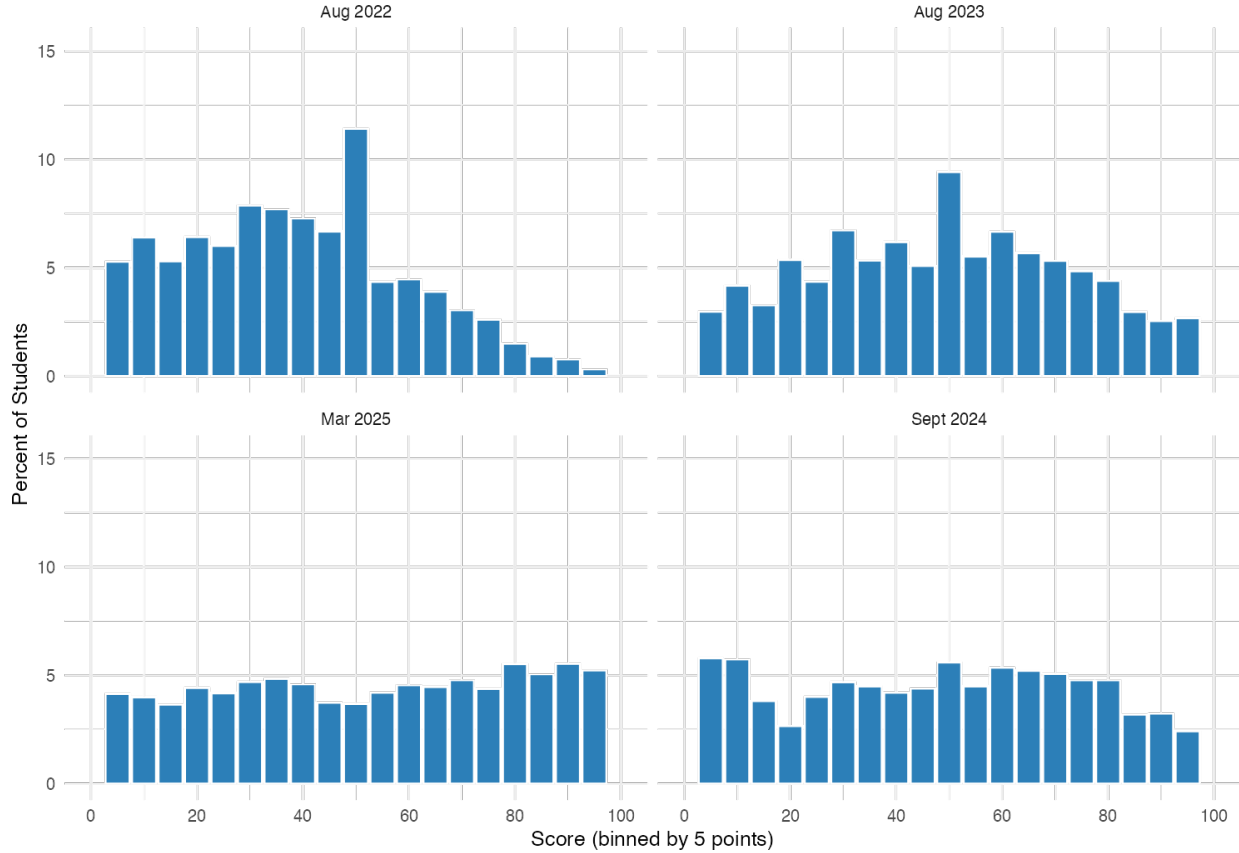


Figure 7: Distribution of school-level Telugu scores across assessment waves.

8 Empirical Strategy

8.1 Overview

This study employs two complementary empirical approaches to examine whether mentor visits are associated with student learning outcomes. The two approaches use different samples and exploit different sources of variation, and together they offer a more complete picture than either would alone.

Approach 1 is a baseline-adjusted school-level regression. It exploits variation in visit intensity across schools within the matched assessment sample, asking whether schools that received more mentor visits between September 2024 and February 2025 achieved higher endline scores in March 2025, conditional on how those schools were performing at baseline.

Approach 2 is a difference-in-differences design. Rather than relying on within-sample variation in visit counts, it exploits variation in the intensity of CSF support across districts,

comparing learning trajectories in the five DPMU districts to those in non-DPMU districts before and after DPMU rollout.

The two approaches answer related but distinct questions. Approach 1 asks whether more visits are associated with better outcomes at the school level. Approach 2 asks whether the broader package of CSF-intensive support, of which more frequent and structured mentoring is one component, is associated with stronger learning gains at the district level.

8.2 Approach 1: Baseline-Adjusted School-Level Analysis

The first approach compares March 2025 endline learning outcomes across schools, controlling for baseline performance to account for pre-existing differences in school quality. To further address the non-random allocation of visits documented above, observable school characteristics are included as additional controls.

The estimating equation is:

$$Y_{s, \text{Mar25}} = \beta_0 + \beta_1 \text{Visits}_s + \sum_r \delta_r \mathbf{1}(\text{round}_s = r) + \sum_r \gamma_r (Y_{s, \text{baseline}} \times \mathbf{1}(\text{round}_s = r)) + \theta_1 \text{Enrollment}_s + \theta_2 \text{Rural}_s + \theta_3 \text{SC/ST}\%_s + \varepsilon_s \quad (2)$$

Here $Y_{s, \text{Mar25}}$ is the school mean Anderson index score in March 2025, and Visits_s is the cumulative number of mentor classroom observation visits received by school s between September 2024 and February 2025. The baseline-year fixed effects δ_r absorb level differences across the three baseline cohorts (2022, 2023, 2024), and the interaction terms γ_r allow baseline performance to differ by round, reflecting the different time gaps between baseline and endline for each cohort. Enrollment_s , Rural_s , and $\text{SC/ST}\%_s$ control for the observable school characteristics shown above to predict visit intensity. Standard errors are clustered at the school level. The model is estimated separately for Mathematics, English, and Telugu.

Approach 1: Results

The baseline-adjusted regression finds no positive association between cumulative visit counts and March 2025 learning outcomes for any of the three subjects. The results are reported in Table 3. For Mathematics, the coefficient on visits is 0.010 (SE 0.014, $p=0.464$), which is near zero and not statistically significant. For English, the coefficient is -0.020 (SE 0.017, $p=0.246$), and for Telugu, the coefficient is -0.009 (SE 0.016, $p=0.577$); neither is statistically significant. The coefficients for English and Telugu are small and negative but

not statistically significant, and likely reflect differences in the types of schools receiving more visits rather than a genuine negative effect of mentoring. Overall, visit counts explain little of the variation in endline outcomes after conditioning on baseline performance and school characteristics.

Table 3: Approach 1: Baseline-Adjusted Regression of March 2025 Scores on Visit Counts

Subject	Coef.	SE	p-value	R^2	N
Mathematics	0.010	0.014	0.464	0.031	297
English	−0.020	0.017	0.246	0.048	290
Telugu	−0.009	0.016	0.577	0.209	246

Notes: School-level baseline-adjusted model. Controls include baseline-year fixed effects, round-specific baseline score interactions, school enrollment, a rural indicator, and percent SC/ST students. Standard errors clustered at the school level.

8.3 Approach 2: Difference-in-Differences

The second approach uses a difference-in-differences design to explore whether schools in DPMU districts experienced larger gains in learning outcomes over time compared to schools in non-DPMU districts. This approach uses a larger sample than Approach 1 because it does not require schools to appear in both a baseline and the endline assessment round. The five DPMU districts are Bhadradi Kothagudem, Medak, Medchal Malkajgiri, Narayanpet, and Vikarabad, with treatment beginning in November–December 2023 for the first three and in September–October 2024 for Narayanpet and Vikarabad, creating staggered treatment timing. The estimating equation is:

$$y_{st} = \alpha + \beta_1 \text{Post}_t + \beta_2 \text{Treat}_s + \beta_3 (\text{Treat}_s \times \text{Post}_t) + \varepsilon_{st} \quad (3)$$

Here y_{st} is the school-wave mean raw subject score, Post_t is an indicator for the March 2025 endline period, Treat_s is an indicator for DPMU district schools, and β_3 is the difference-in-differences estimator which captures the differential change in learning outcomes in DPMU districts relative to non-DPMU districts. The model is estimated using August 2023 as the baseline period and March 2025 as the endline. Standard errors are clustered at the school level throughout.

Approach 2: Results

Figure 8, 9, and 10 present visual evidence for the difference-in-differences regressions in Mathematics, English, and Telugu. In each figure, the purple line shows mean raw scores for DPMU districts, while the green line shows mean raw scores for non-DPMU districts across assessment waves. The shaded purple region marks the period following the initial rollout of CSF-intensive support in Bhadradri Kothagudem, Medak, and Medchal Malkajgiri in November–December 2023, while the shaded red region marks the later expansion to Narayanpet and Vikarabad in September–October 2024.

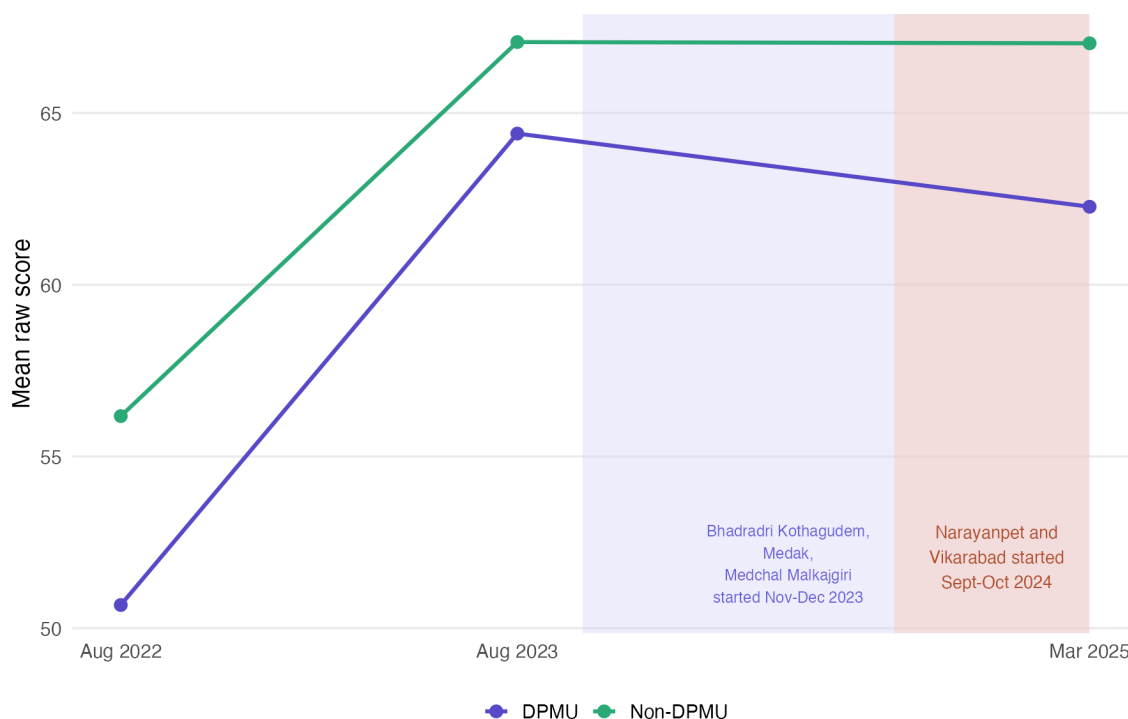


Figure 8: Difference-in-differences: Mathematics. DPMU vs. non-DPMU districts across assessment rounds.

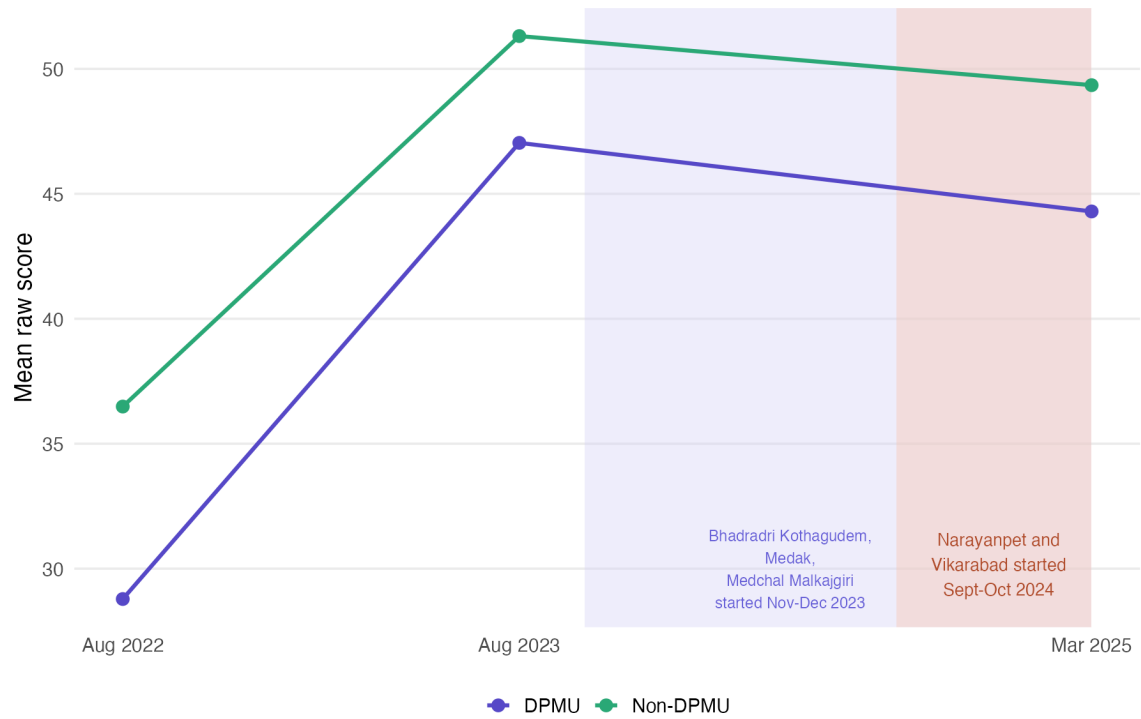


Figure 9: Difference-in-differences: English. DPMU vs. non-DPMU districts across assessment rounds.

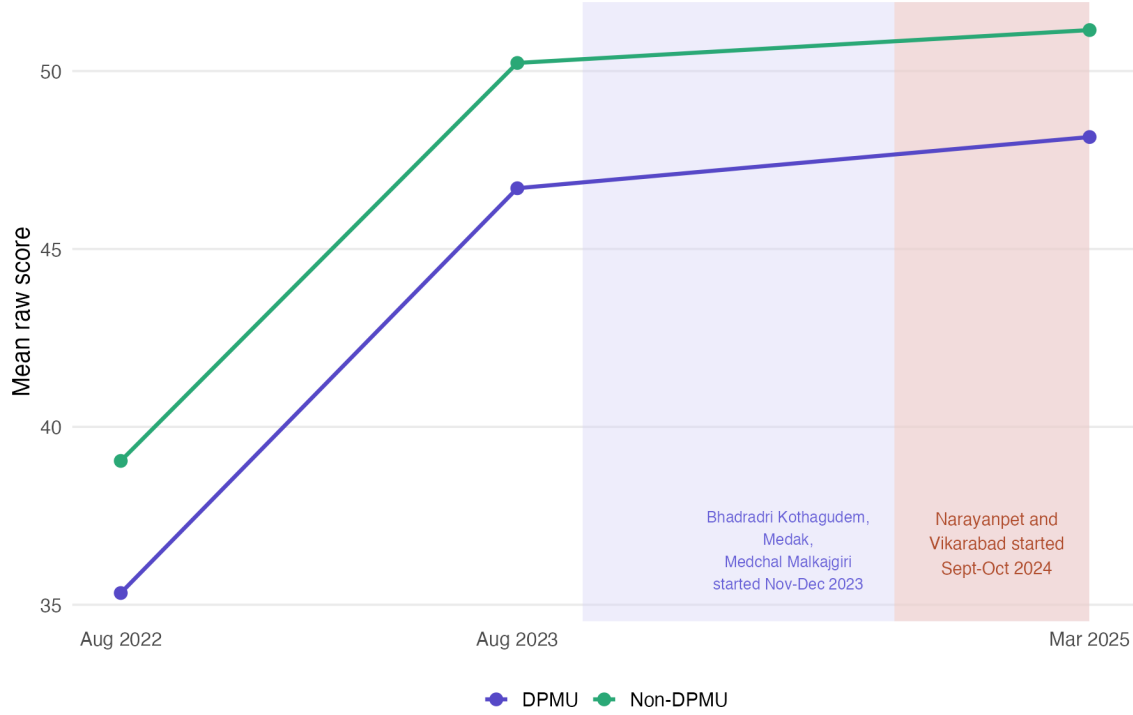


Figure 10: Difference-in-differences: Telugu. DPMU vs. non-DPMU districts across assessment rounds.

Table 4 reports the corresponding regression results. The coefficient of interest is the interaction term, $\text{Treat} \times \text{Post}$, which captures whether DPMU districts improved more than non-DPMU districts over time. Across all three subjects, the interaction term is statistically insignificant: -2.094 in Mathematics, -0.780 in English, and 0.516 in Telugu. The Treat coefficient is negative and significant across all three subjects, confirming that DPMU schools scored lower than non-DPMU schools at baseline in August 2023, consistent with the programme targeting lower-performing districts. The insignificant interaction term indicates, however, that no differential improvement relative to non-DPMU districts is detected by March 2025. Notably, the Post coefficient for English is negative and significant (-1.964), suggesting a general decline in English scores over this period across both groups. Overall, there is no evidence that DPMU districts experienced differential learning gains relative to non-DPMU districts over this period.

Table 4: Approach 2: Difference-in-Differences Results, August 2023 vs. March 2025

	Mathematics	English	Telugu
Post	−0.0358 (0.7370)	−1.9643** (0.9105)	0.9241 (0.9020)
Treat	−2.6634** (1.2520)	−4.2736*** (1.4042)	−3.5215** (1.4255)
Treat × Post	− 2.0944 (1.8651)	− 0.7800 (2.2129)	0.5155 (2.3195)
R^2	0.0054	0.0071	0.0032
N	3103	3009	2881

Notes: Each column reports a separate difference-in-differences regression using school-wave mean raw subject scores. The omitted group is non-DPMU schools in August 2023. **Post** indicates March 2025, **Treat** indicates schools in the 5 DPMU districts, and **Treat × Post** is the coefficient of interest. Standard errors clustered at the school level are shown in parentheses. Treatment started in Nov–Dec 2023 for Bhadradi Kothagudem, Medak, and Medchal Malkajgiri, and in Sept–Oct 2024 for Narayanpet and Vikarabad. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9 Discussion

Both empirical approaches find no robust positive association between mentor visit exposure and student learning outcomes during the study period. The baseline-adjusted regression results show near-zero or negative coefficients across all three subjects, and the difference-in-differences analysis finds no consistent differential gains in learning outcomes in DPMU districts relative to non-DPMU districts. These results might be read as evidence that the mentoring pathway is ineffective. However, we argue that limitations in our data, necessary assumptions we had to make for the analysis, and the timing of treatment and outcome measurement make it difficult to detect a true effect even if one exists.

The most direct explanation for the null result is that the study window is too short to observe the full causal chain implied by the theory of change. The mentoring pathway works through a sequence of steps: mentors visit classrooms, observe teaching, and provide feedback to teachers who then internalize that feedback and change their teaching practices. Students eventually respond to improved teaching with measurable learning gains. Each link

in this chain takes time. Cumulative visits are measured over six months, and the outcome is assessed just one month after the treatment window closes. Therefore, this study design may not allow sufficient time for this process to unfold.

A second explanation concerns aggregation assumptions that we had to make due to limitations in the data. In practice, a recorded mentor visit involves the observation of a single classroom. However, in order to carry out our analysis at a level to match both student assessment and teacher visits, we aggregated all data to the school level. Therefore, the outcome variable is the school-level mean score aggregated across all assessed students in that particular school. The students who sit the assessment may not be those taught by the observed teacher, and the subject observed during the visit may not correspond to the subject being assessed. This mismatch between the unit of treatment and the unit of measurement introduces potential attenuation bias that may push estimated coefficients toward zero even if there was a positive effect. The small and statistically insignificant coefficients in Approach 1 are therefore difficult to interpret as evidence against effectiveness of the intervention since the study design cannot isolate the students or classrooms directly affected by the intervention.

A third and related concern is the distinction between visit quantity and visit quality. This study measures only the compliance dimension of mentoring, i.e., how many visits occurred. It does not capture whether the feedback delivered during those visits was substantive or actionable. The concern about top-coding in the observation instrument - whereby mentors may select the highest response categories to avoid conflict rather than to accurately reflect classroom practice - suggests that even the recorded content of visits may not reflect genuine engagement.

It is worth distinguishing this from the data limitations above. The first two concerns suggest that visit frequency may cause improved learning, but that this study is not well-positioned to detect it. This third concern raises a different possibility: that visit frequency may not cause improved learning at all, if it is the quality rather than the quantity of visits that matters. Even a well-designed study with perfectly matched units of observation might find a null result if the active ingredient is the depth and quality of mentor feedback rather than the number of visits recorded.

The DiD design offers a different angle on the same question, using variation in CSF support intensity across districts, rather than variation in visit counts. However, the results in this approach are similarly inconclusive. Two of the five DPMU districts (Narayanpet and Vikarabad) only joined the programme in September-October 2024, meaning they were exposed to CSF-intensive support for at most five or six months before the March 2025 end-

line. For these districts, the DiD comparison is effectively measuring the immediate impact of a very recent intervention. The sign instability of the DiD coefficients across two baseline comparisons (August 2022 and August 2023, with the former examined as a robustness check and not reported) further suggests the estimates are sensitive to pre-treatment trend assumptions, which limits confidence in any single comparison. The DiD results should therefore be read as exploratory rather than as a definitive test of whether DPMU-level support improves learning.

One important and actionable finding of this study is not in the regression results but in the descriptive analysis of visit distribution. We show that larger and more urban schools systematically receive more mentor visits, while smaller, more rural, and schools with a higher proportion of low-caste students receive fewer visits. From a program equity consideration, this implies that the schools most likely to need structured support are the least likely to receive it. If mentoring does have positive returns at the margin, concentrating visits on schools that are urban and larger, and likely already better-resourced, will produce smaller system-level gains than distributing visits more equitably. More importantly, if the reform is intended to narrow foundational learning gaps, a visit allocation pattern that mirrors existing inequalities risks reinforcing rather than reducing them. The narrowing of the urban–rural gap between 2024/25 and 2025/26 is an encouraging sign that visit distribution is improving. However, the underlying targeting pattern remains a concern.

Taken together, the findings do not provide evidence that the mentoring pathway has improved student learning outcomes during the study period, but neither do they provide evidence that it has not. The data and design constraints, including non-random visit allocation, short treatment window, aggregation mismatch (between teachers, students and schools) for the analysis, and limited sample size, are collectively important limitations of this study.

10 Conclusion

This study set out to examine whether variation in mentor visit exposure and CSF support intensity is associated with differences in student learning outcomes in Telangana. We find across both empirical approaches and all three subjects, that there is no robust positive association in the current data. However, given the short treatment window, aggregation constraints, non-random visit allocation, and limited sample, this finding should be read as a measurement challenge rather than a verdict on program effectiveness.

The clearest and most actionable finding is distributional. Mentor visits are systematically concentrated in larger, more urban, and less disadvantaged schools. This pattern is improving: the urban–rural gap narrowed between 2024/25 and 2025/26, but the underlying targeting logic remains a structural concern. A reform designed to narrow learning gaps cannot achieve that goal if the schools most in need of support are the least likely to receive it. Addressing this requires not just higher aggregate visit targets, but changes to how compliance is monitored and how mentors are held accountable for equitable coverage.

Future research should prioritize three things. First, a longer follow-up window would allow the theory of change sufficient time to operate. This would ideally be two or more full academic years of visits linked to subsequent outcome data. Second, improved measurement of visit quality that goes beyond compliance counts. This is essential for understanding whether the content of mentoring is changing classroom practice. Third, school-level linkage between visited classrooms and assessed students would resolve the aggregation mismatch that currently makes it impossible to isolate the students most directly exposed to the intervention.

While this study cannot conclusively determine whether mentor visits are improving foundational learning in Telangana, the conditions for the mentoring pathway to succeed are not yet fully in place. Visit frequency is improving but remains below the six-visit benchmark for many schools, the quality of feedback delivered during visits cannot be reliably measured or studied, and the distribution of visits continues to favor larger and more urban schools. Until these foundations are stronger, and until longer-term outcome data are available to detect effects that may take more than one academic year to materialize, the reform’s impact on student learning will remain difficult to assess with confidence.

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